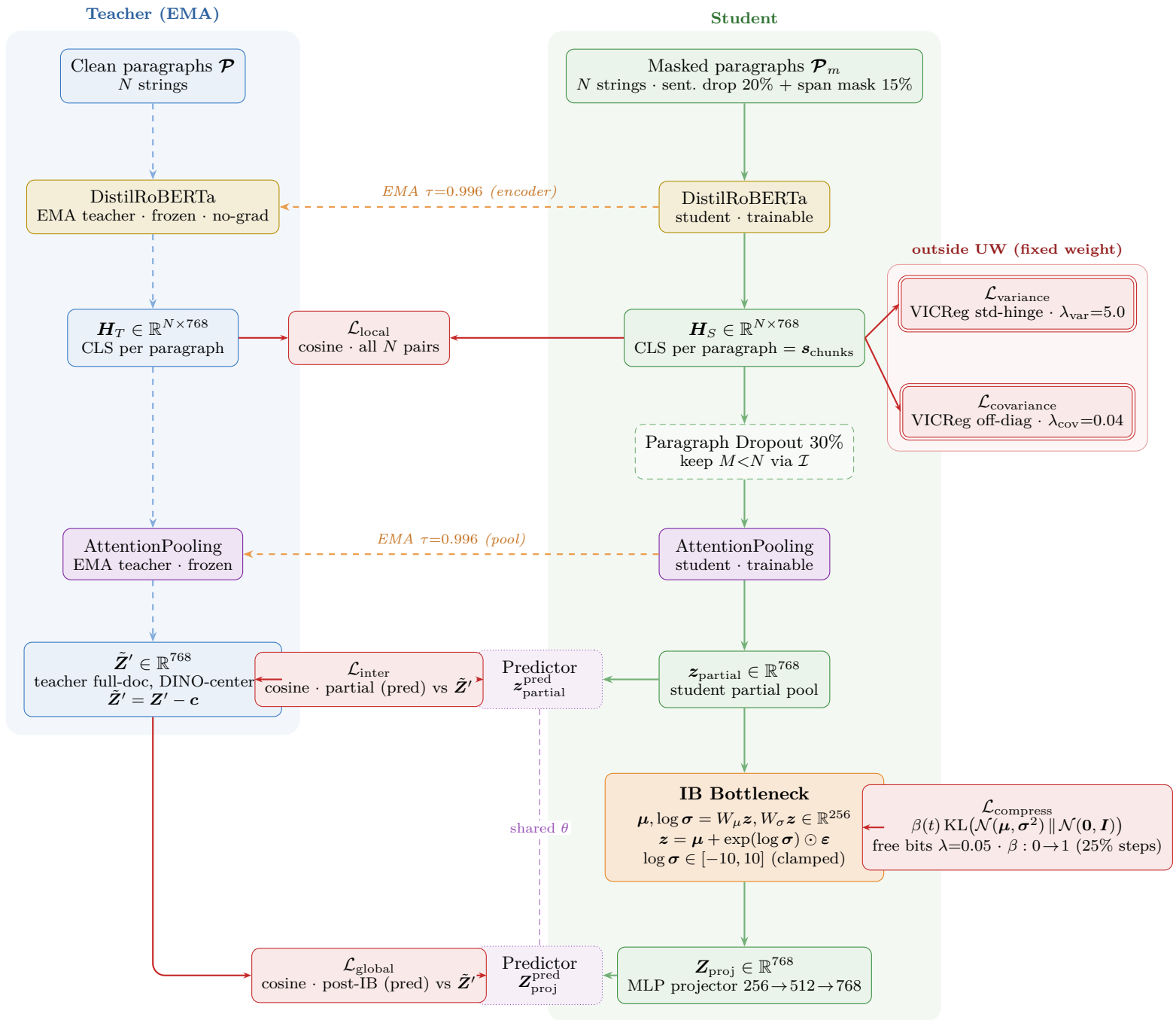




Total loss — Uncertainty-Weighted alignment + fixed-weight VICReg

$$\mathcal{L} = \sum_{i \in \{\text{compress}, \text{local}, \text{inter}, \text{global}\}} (e^{-s_i} \mathcal{L}_i + s_i) + \lambda_{\text{var}} \mathcal{L}_{\text{variance}} + \lambda_{\text{cov}} \mathcal{L}_{\text{covariance}}$$

$s_i = \log \sigma_i^2$ learnable, clamped to $[-10, 10]$ $\lambda_{\text{var}}=5.0, \lambda_{\text{cov}}=0.04$ fixed (never down-weighted by UW)

— trainable (student) — EMA teacher (no-grad) — EMA update (encoder + pool), $\tau=0.996$ — loss signal

[...] Predictor: student-only BYOL/SimSiam head 768 → 512, LN, GELU, → 768 (shared θ ; teacher has none) ▮ double border = fixed weight, outside UW

Training also applies temporal VICReg on $\mathbf{Z}_{\text{proj}}^{\text{pred}}$ and μ via rolling buffers (script-level regulariser; not shown).